HOMEWORK 8.3 · UNIT 8 · NC.8.G.3 · NC.8.G.4 · NC.8.G.2

Dilations (Intro)

Apr 12 / Apr 13 · Targets: G.3-4, G.4-1, G.3-3, G.2-3, G.3-5, G.3-7, G.3-9, G.2-9

Name _____ Date _____ Block _____

Remember: Dilation about the origin with scale factor k: $(x, y) \to (kx, ky)$. Multiply **both** numbers by k. $k > 1$: bigger (enlargement). $0 < k < 1$: smaller (reduction). Angles stay the same, so the image is **similar** (~), not congruent.
Example: $k = 3$: $(2, 1) \to (3 \times 2, 3 \times 1) = (6, 3)$.

SCORE 3.0 Approaching · the basics

Targets: G.3-4, G.4-1, G.3-3, G.2-3

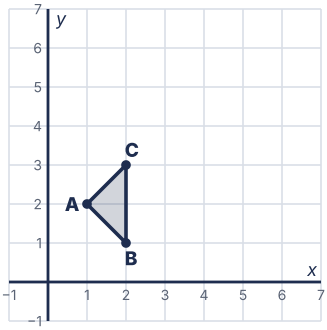
1 G.3-4

△MNP has M(−4, 4), N(6, 2), P(2, −6). Dilate it by $k = \frac{1}{2}$ about the origin. Find N′.

Ⓐ (3, 1) Ⓑ (12, 4) Ⓒ (6.5, 2.5) Ⓓ (3, 2)

2 G.3-4

Dilate △ABC by $k = 2$ about the origin. A(1, 2), B(2, 1), C(2, 3). Write A′, B′, C′. Draw the image.



3 G.4-1

A dilation has $k = 0.75$. Circle one.

Enlargement Reduction

Why? _____

4 REVIEW 8.2 G.3-3

Rotate (5, −3) 90° counterclockwise about the origin.

Ⓐ (−3, −5) Ⓑ (−5, 3)
Ⓒ (3, 5) Ⓓ (5, 3)

5 REVIEW 8.1 G.2-3

Translating a figure 3 units up changes its shape.

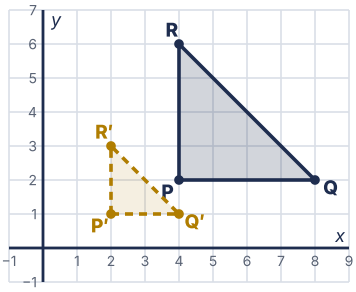
True False

SCORE 4.0 Proficient · on target

Targets: G.3-5, G.3-7, G.4-1

G.3-5

6 △PQR → △P′Q′R′ by a dilation about the origin. P(4, 2) → P′(2, 1), Q(8, 2) → Q′(4, 1). Find the scale factor.



G.3-7

7 **Omar** made a mistake. Circle it. Explain it. Fix it.

OMAR'S WORK

Dilate $(6, -3)$ by $k = \frac{1}{3}$.

$(6 \times 3, -3 \times 3) = (18, -9)$

G.4-1

8 A triangle has a 10 cm side. It is dilated by $k = \frac{1}{2}$. How long is the new side?

Ⓐ 5 cm Ⓑ 20 cm Ⓒ 10.5 cm Ⓓ 9.5 cm

SCORE 5.0 Mastery · go deeper

Targets: G.3-9, G.2-9

G.3-9

9 A square with 4-inch sides is dilated by $k = 3$. Is the image congruent or similar? Explain with side lengths.

The new sides are _____ . The image is _____ because _____ .

G.2-9

10 Explain the difference between a rigid transformation and a dilation. Give one example of each.

A rigid transformation _____ . A dilation _____ .